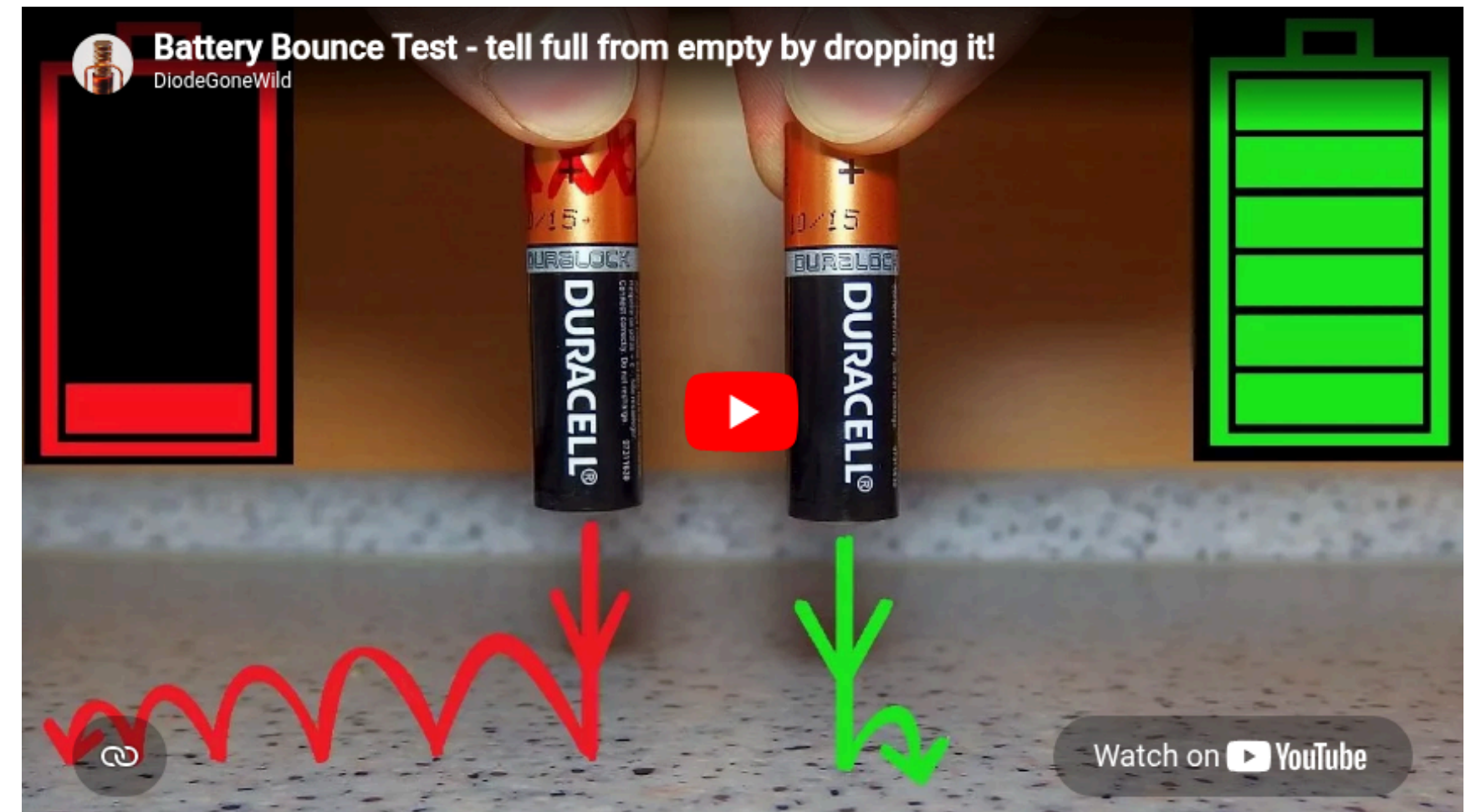


KINETIC VOLTMMETER

Team India

Problem Statement

“A DIY method of testing the charge of a battery is to drop it on a hard surface and observe whether it bounces. Explain the phenomenon and find other non-invasive mechanical methods to estimate the remaining charge. Optimize the accuracy of your method.”

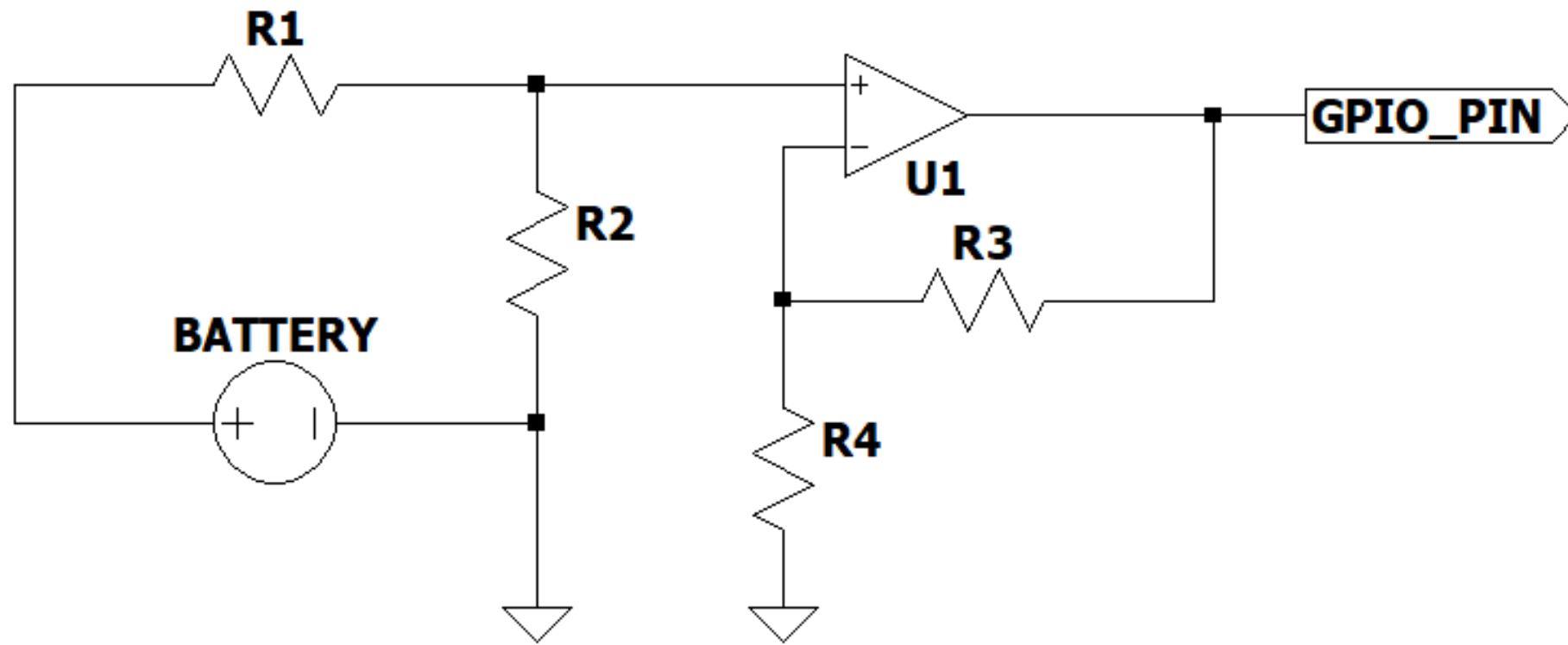


YouTube: DiodeGoneWild

Part I

Do discharged batteries bounce higher?

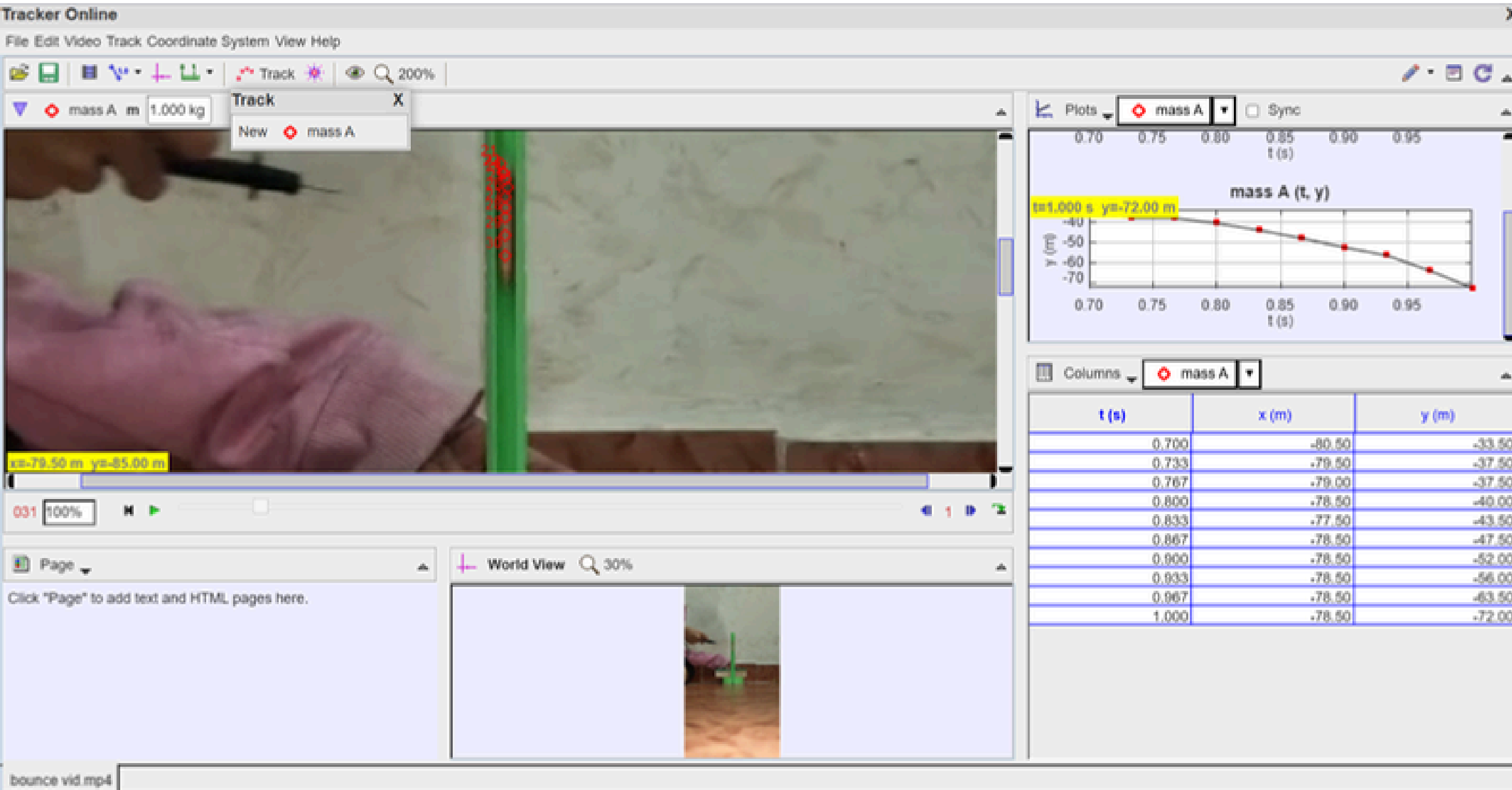
Experimental Setup



Object	Specification
Battery	Duracell AAA Alkaline, 1.5 V
R_1	3.9 Ω , 5% tolerance, 0.5W ceramic resistor
Shunt Resistor R_2	75 m Ω , 5% tolerance
Microcontroller	ESP32-WROOM-32, 240 MHz
DAQ	NI USB-6009, 14-bit
3D Vertical Stand	24 cm height
Phone Camera	480 FPS



Experimental Analysis



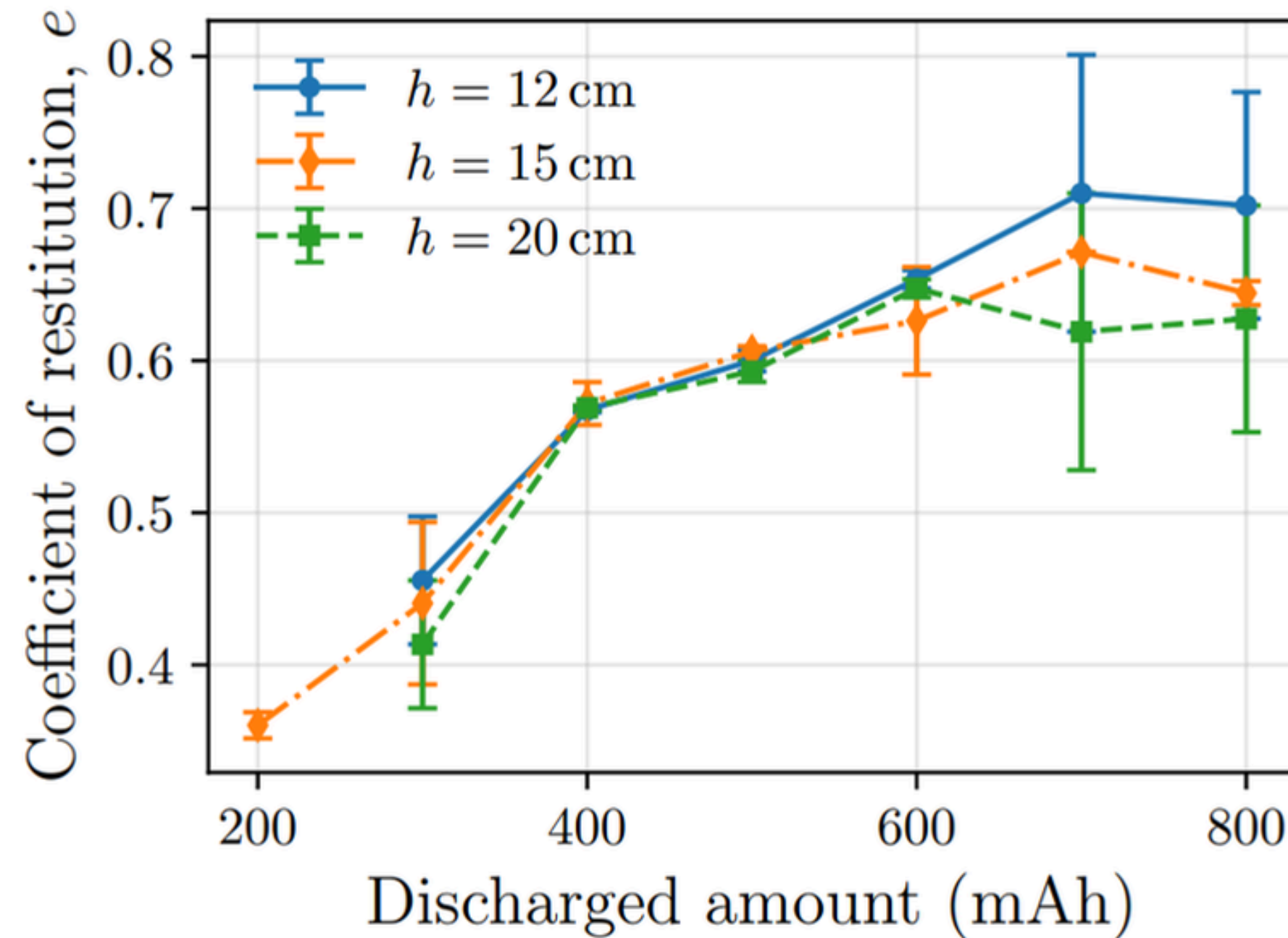
Drop from height h

Track till maximum height h'

$$e = \sqrt{\frac{h'}{h}}$$

Results

Discharged batteries bounce higher!

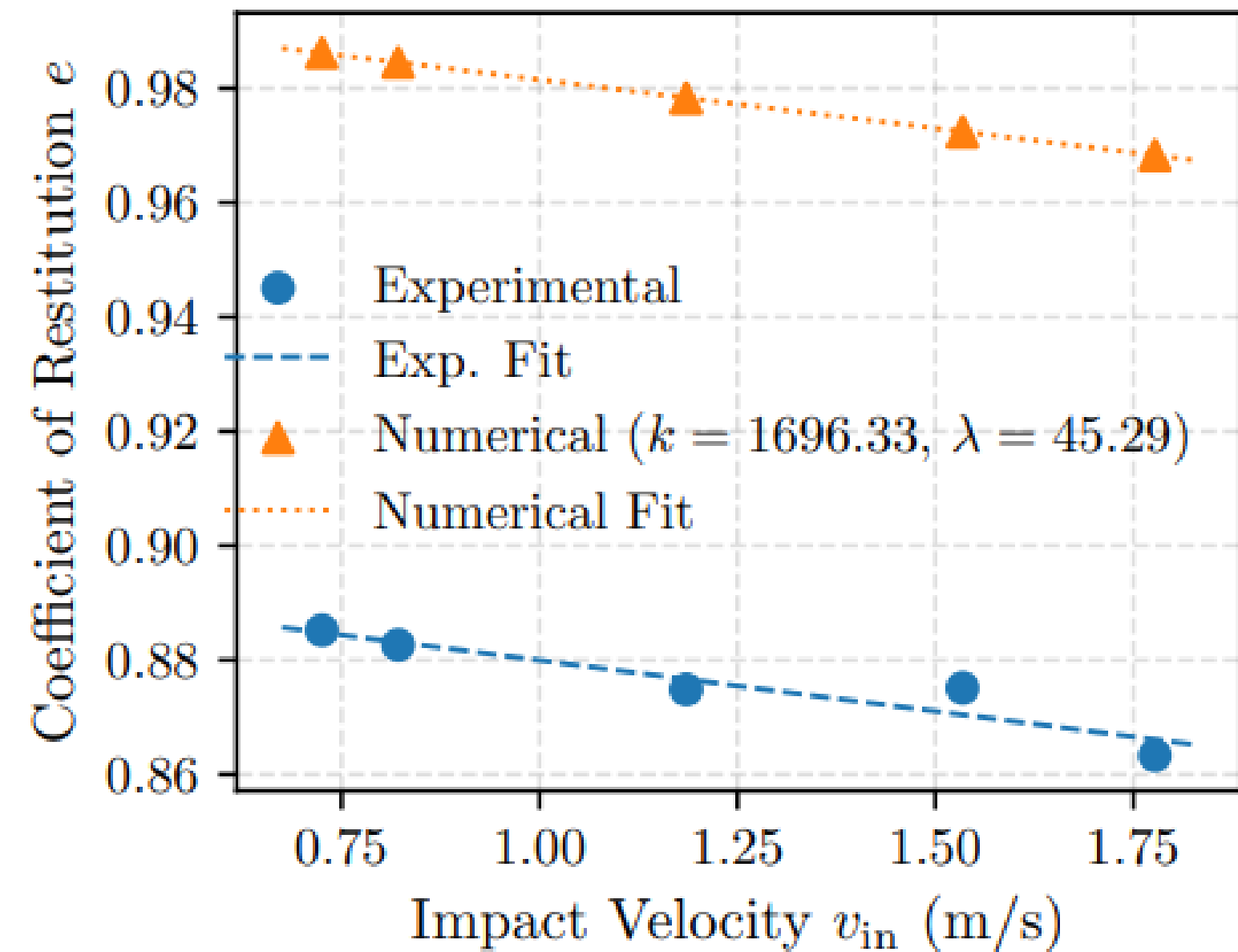
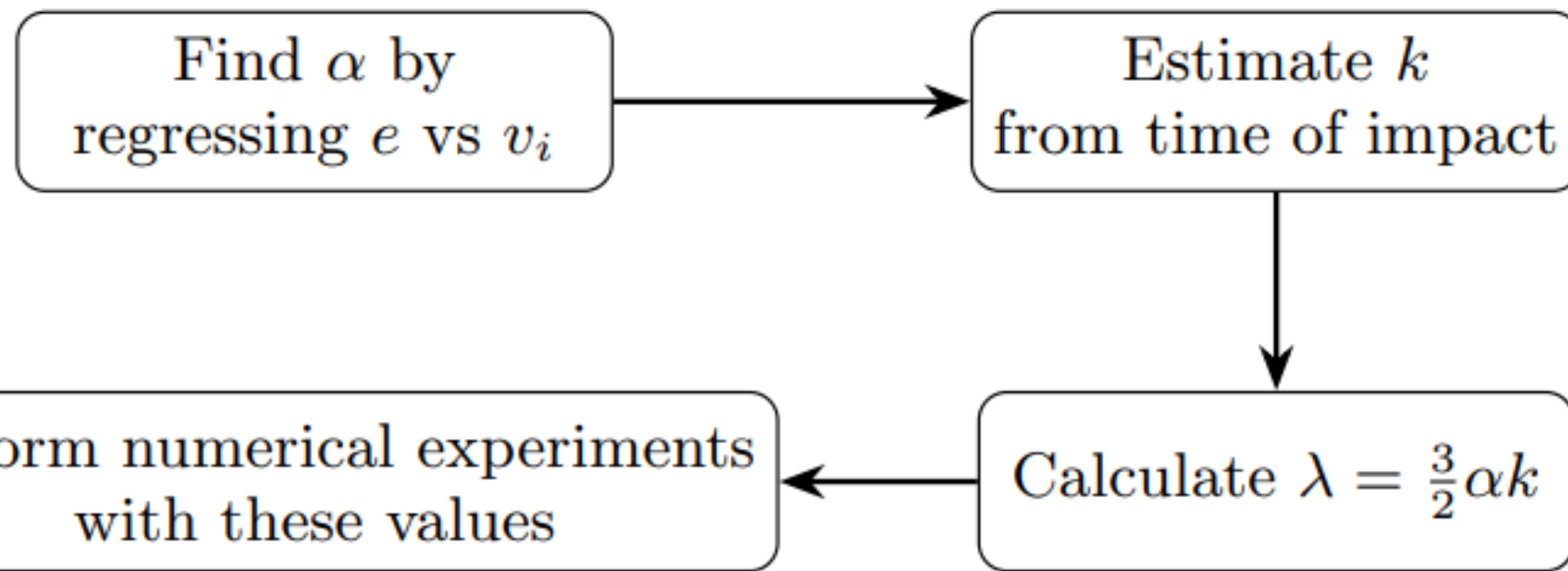


- Nonlinear, complex variation
- COR is smaller for higher speeds, on average

Contact Dynamics

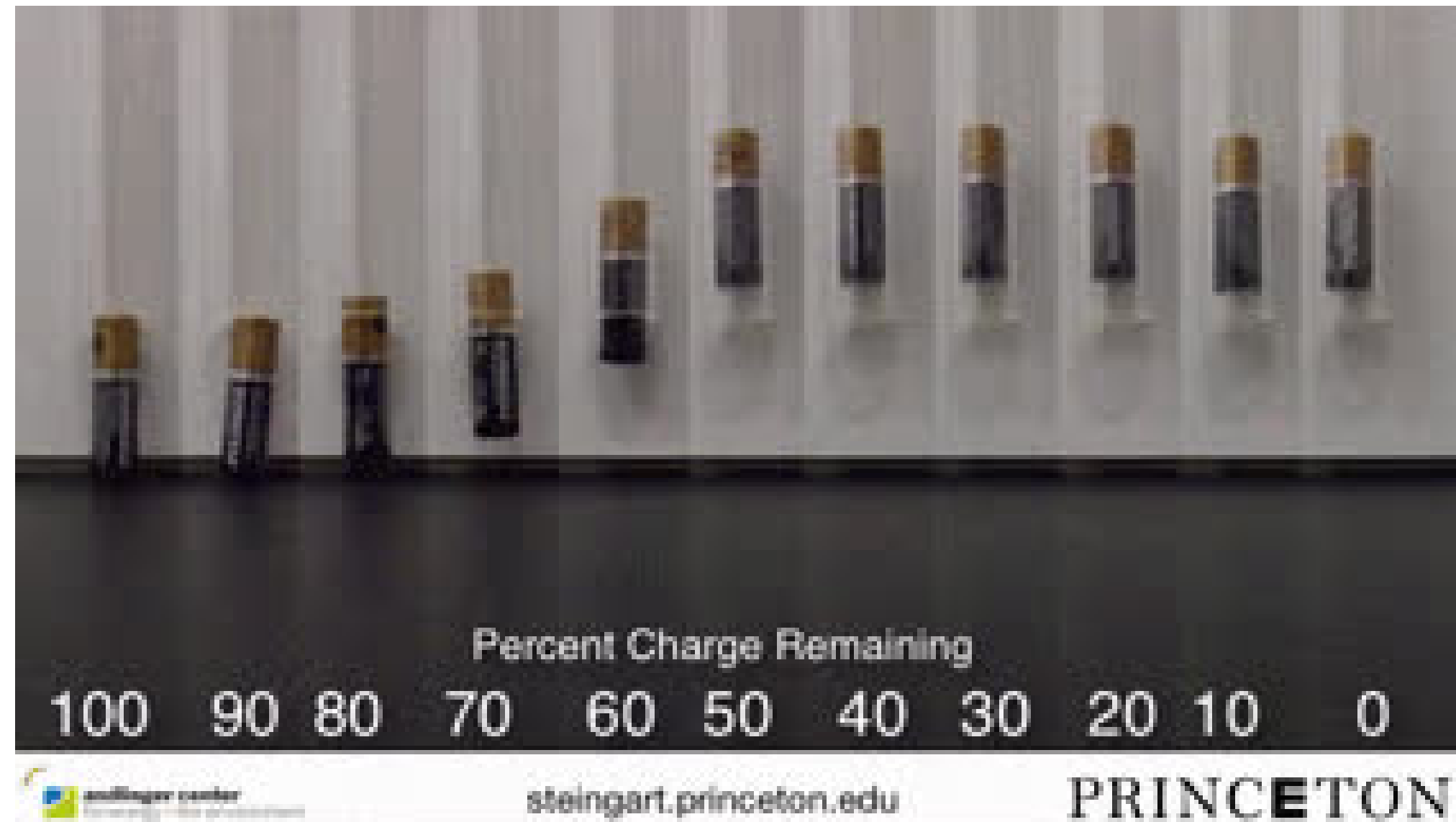
Hunt
Crossley
Model

$$\begin{cases} e \approx 1 - \alpha v_i \\ F_{spring} = -k\delta - \lambda\delta\dot{\delta} \end{cases}$$



What is the reason of the
discrepancy?

An observation...

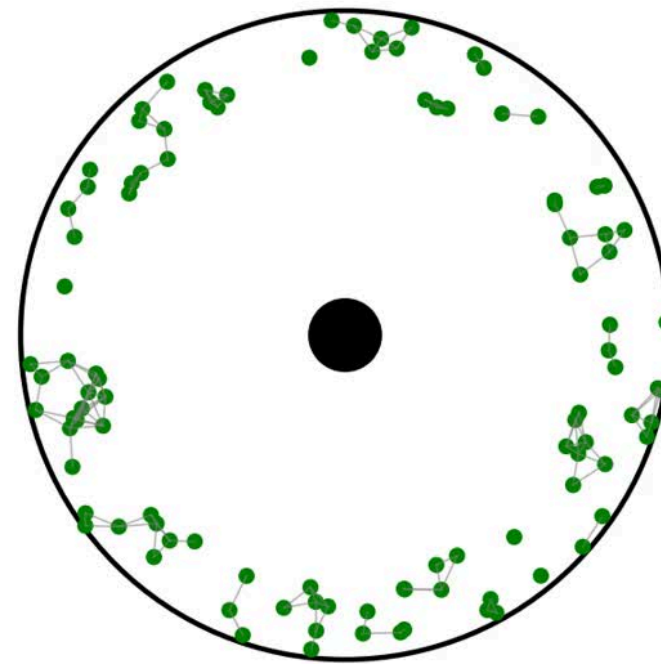
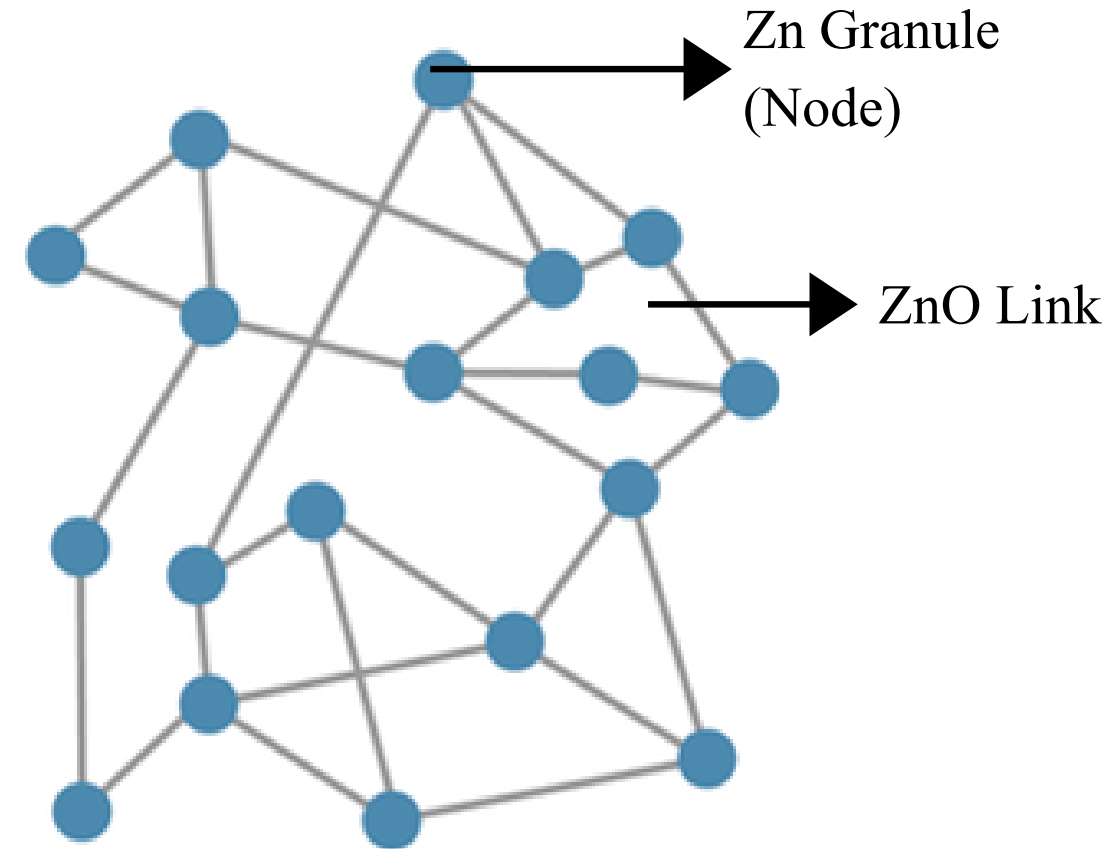
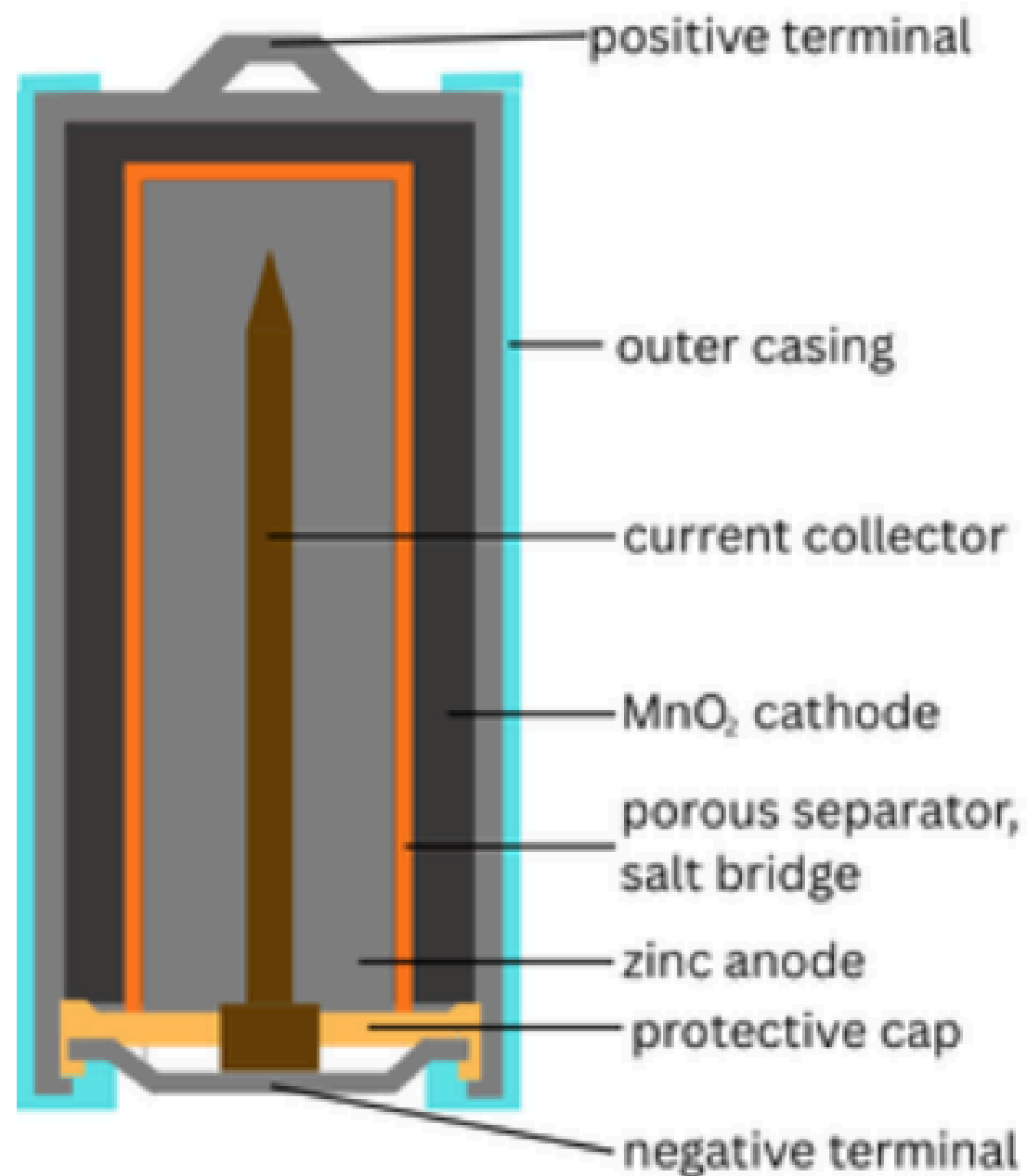


<https://www.princeton.edu>, Bhadra et al., 2015

Average bounce height abruptly jumps at a **critical level of discharge?**

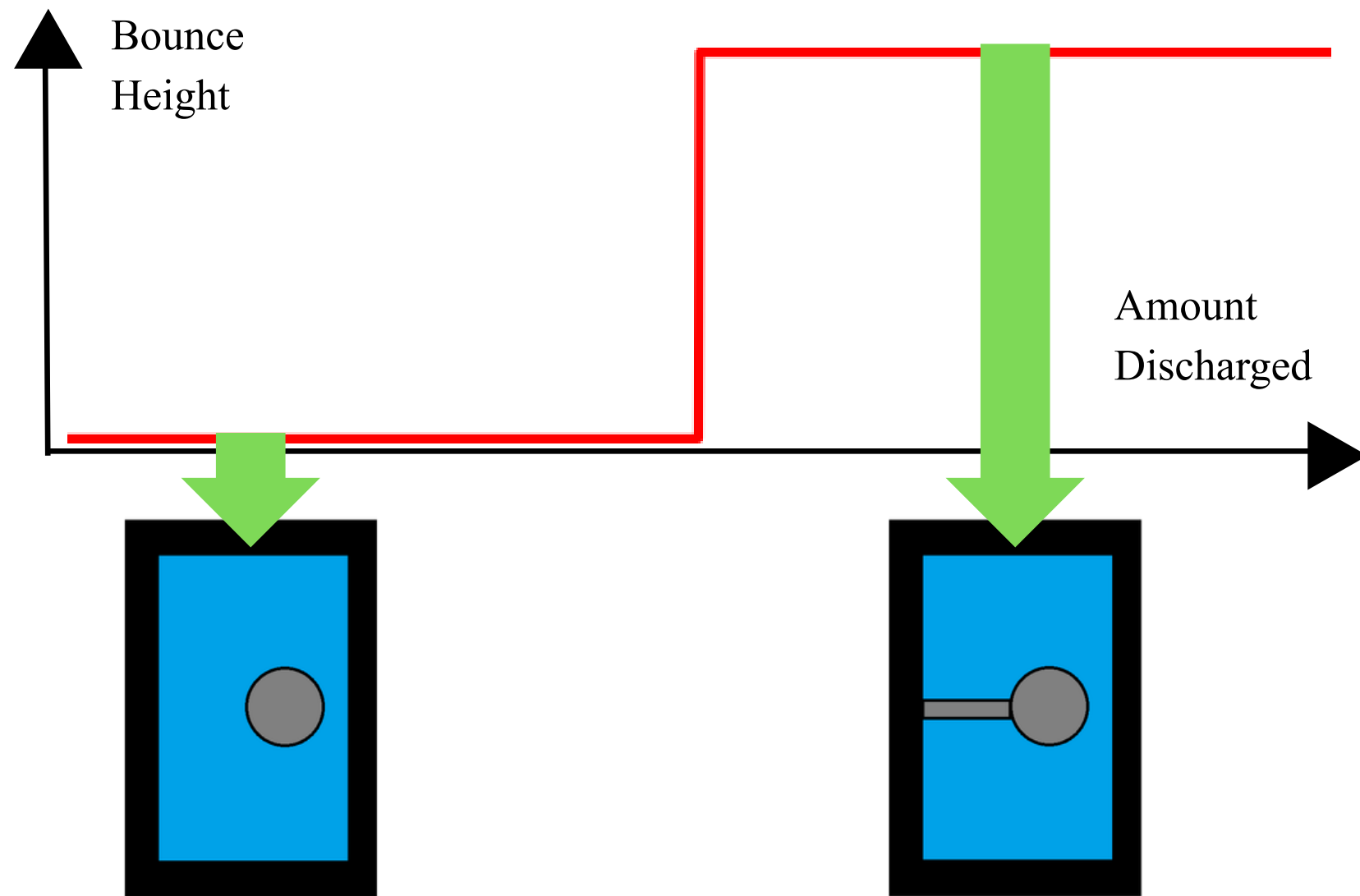
Zn/ZnO Networks

quantamagazine.org

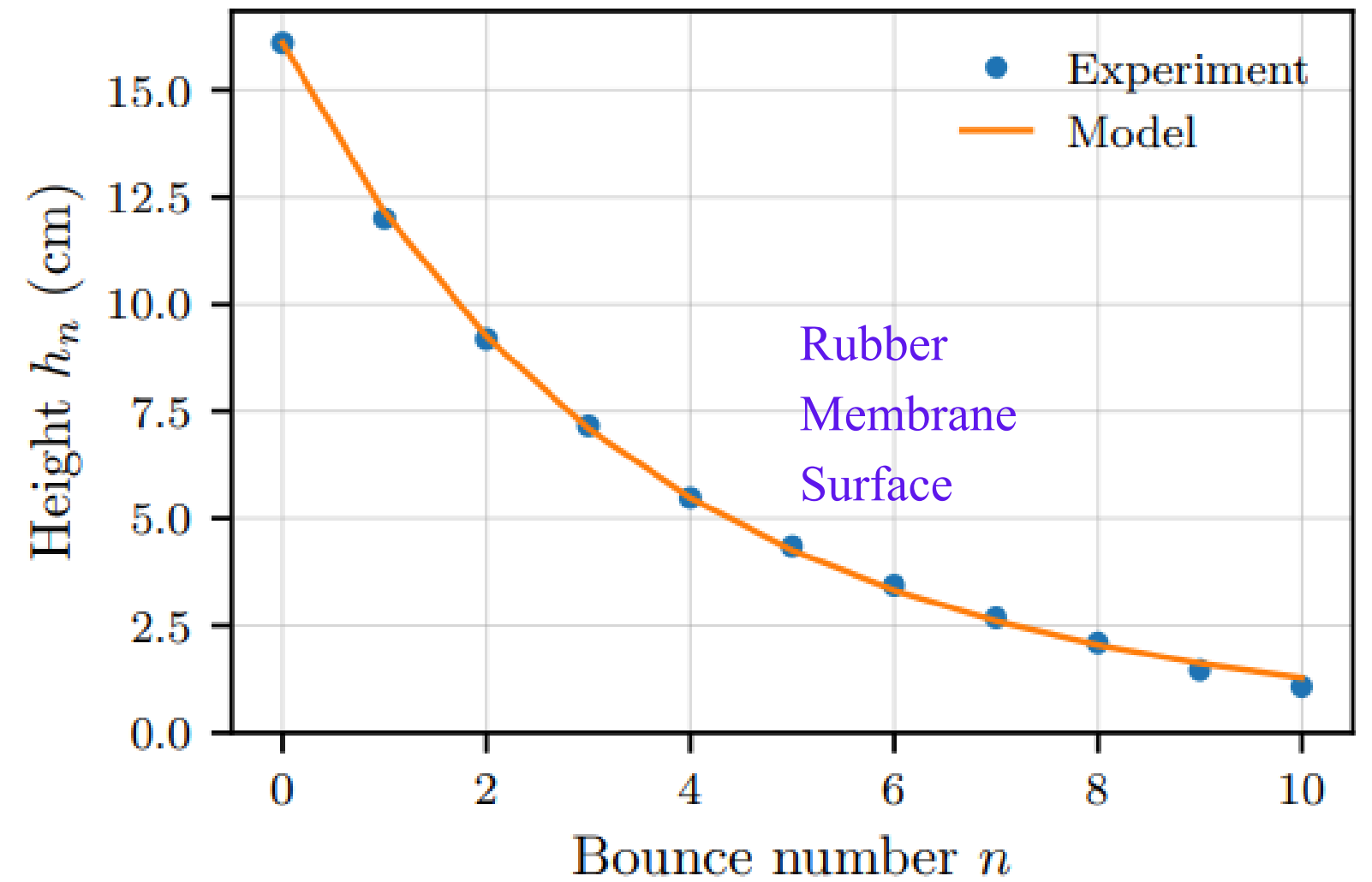


- Pre-contact: Battery not discharged enough for components to reach collector
- Post-contact: Giant component touch collector
- Qualitatively different bouncing behaviour between pre-contact and post-contact

Energy Loss



- 2 channels for energy loss: **impact channel** and **viscous dissipation channel**.
- Viscous dissipation channel disappears once a critical level of discharge is crossed.



$$h_{n+1} = \left[(1 - X) \sqrt{1 - 2\alpha \sqrt{2gh_n}} - X \right]^2 h_n$$

$$X : \frac{\text{Mass of Zn and ZnO in anode}}{\text{Mass of cell}}$$

Part II

Non-Invasive Method

Ultrasonic Testing

Method

Pulse Echo UT method was used.

Transducer emits a 5 MHz pulse into the battery and receives reflections from internal material boundaries.

Hypothesis

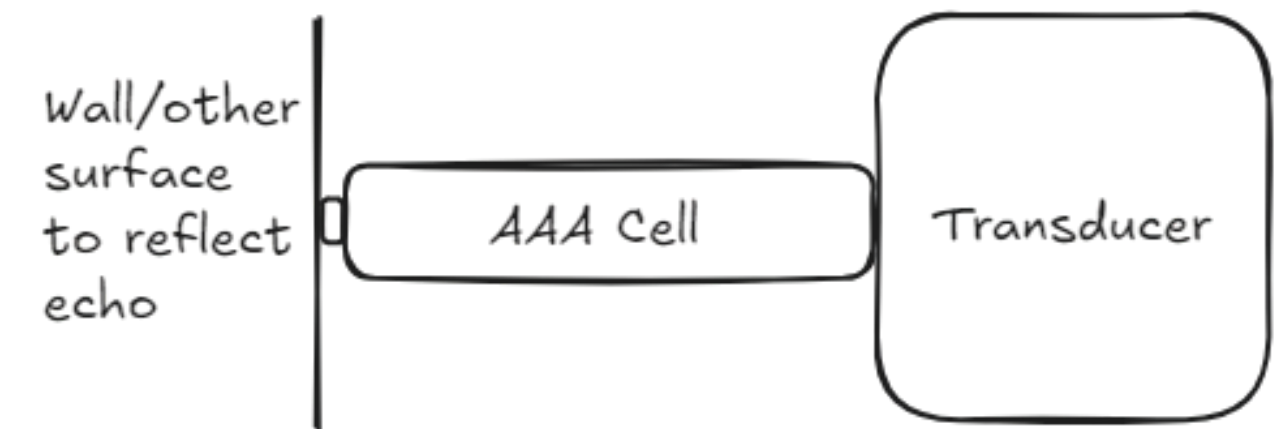
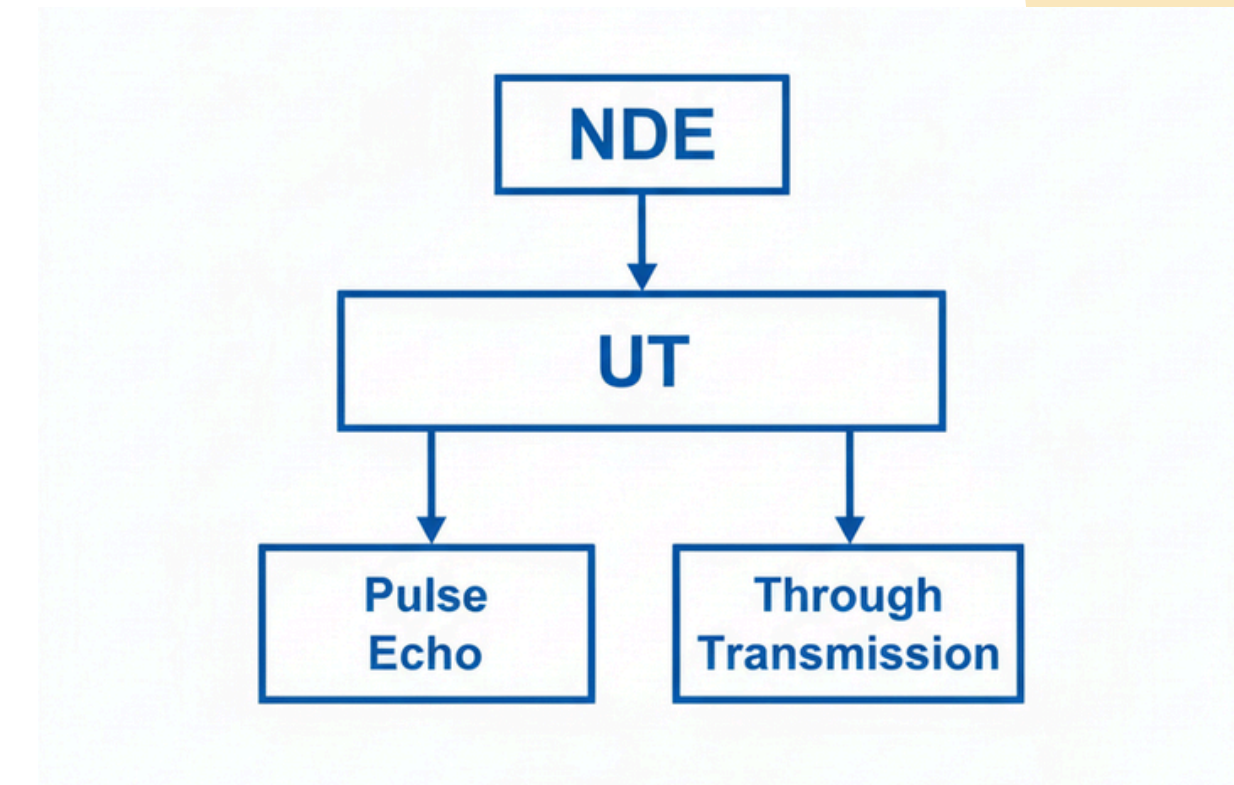
Change in chemical properties → Change in mechanical properties → Change in time of flight and signal attenuation

Scaling Equation

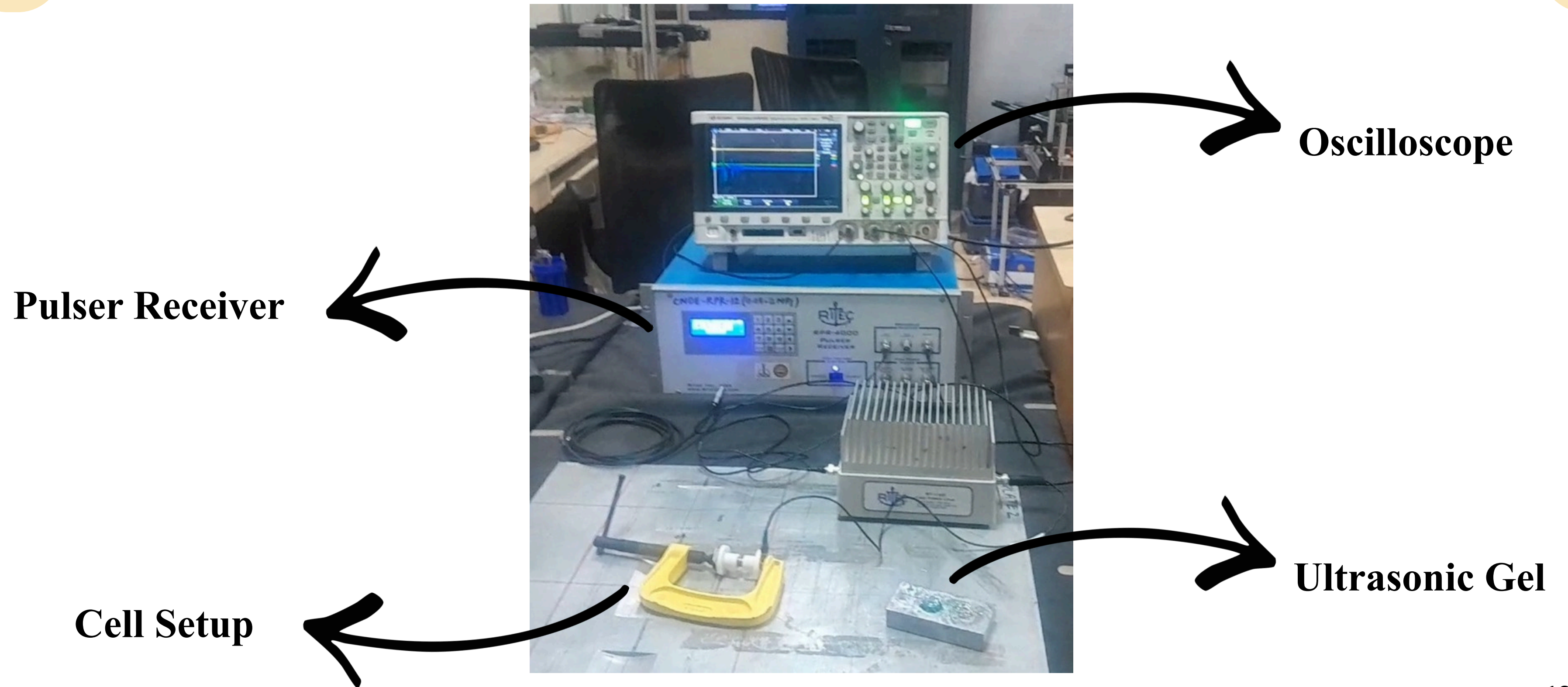
$$G \sim v^2$$

$G \rightarrow$ elastic modulus

$v \rightarrow$ speed of sound



Apparatus



Procedure

Make Connections

Transducer → Pulser Receiver (PR) → Oscilloscope (OS)

Apply UltraSonic gel

to transducers to ensure proper contact

Arrange the setup

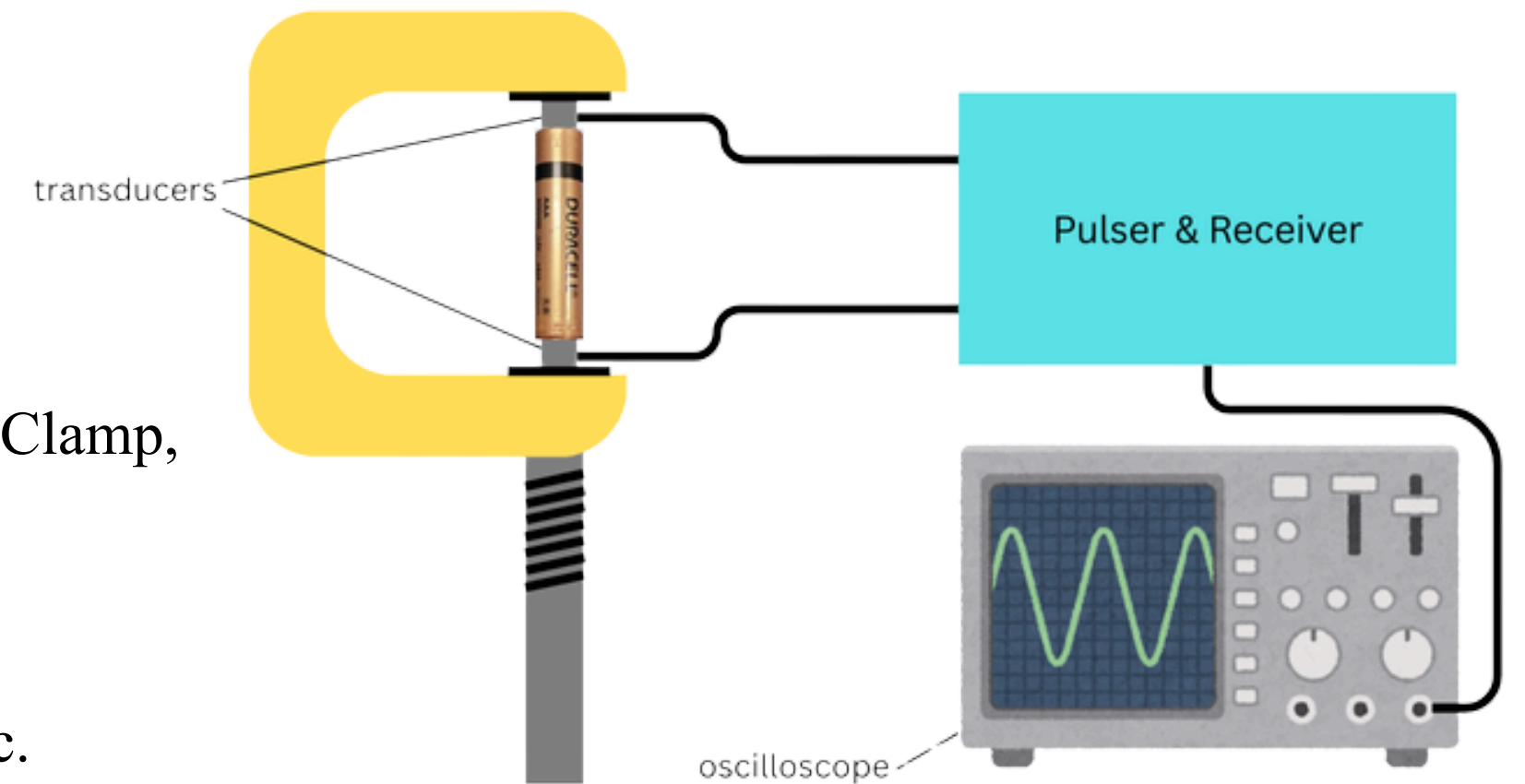
Fix cell and transducer using C-Clamp, and custom clamping 3D print

Configure PR

5MHz, set LP and HP filters, etc.

Download OS Data

And analyze the signal



Results

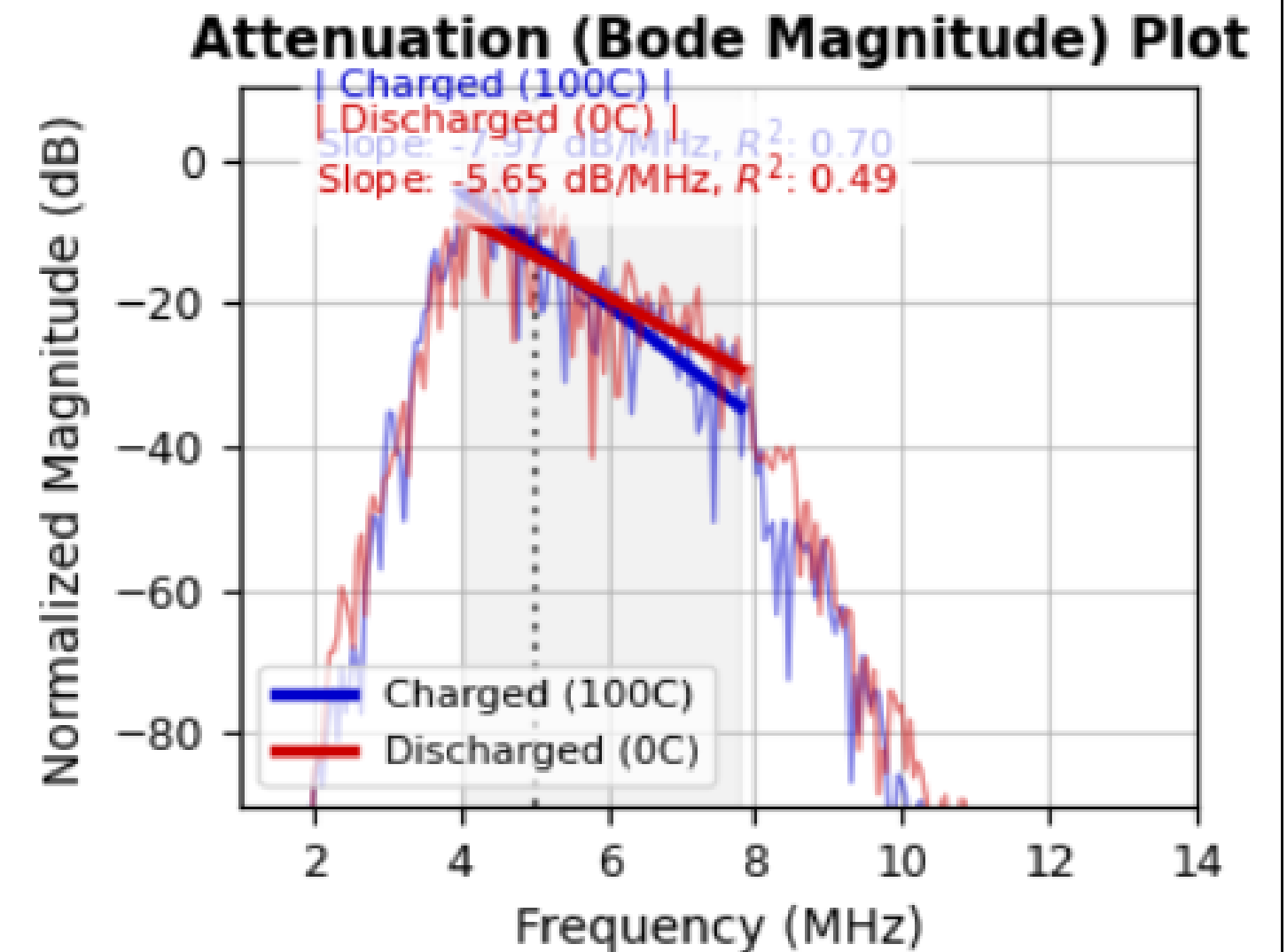
Time of Flight Analysis

Sound will travel slower in a gel-like medium than in rigid structures

Q) How does TOF vary with charge? Smooth?
Nonsmooth?

Acoustic Metric	Charged (100C)	Discharged (0C)
Primary TOF (μ s)	9.52	7.79
Peak Amplitude (V)	0.16	0.67

Attenuation Slopes



Conclusions

Part I

- Discharged batteries bounce higher!
- Hunt-Crossley dynamics does not capture full picture
- Conjectured a critical shift in bounce height
 - possibly signalling towards a phase transition!

Part II

- Pulse Echo UT as a non-invasive method
- Ultrasound definitively travels slower in charged batteries
- Attenuation slopes were quantified and can be explored as an alternative measure of SOC.